Automated RDBMS-to-Knowledge Graph Conversion Bot

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Applied Project

Presentation in fulfillment of requirements of the NYU SPS course

Applied Project: Capstone

Spring, 2026

Project Sponsor and Project Goal

- Project Name: Automated RDBMS-to-Knowledge Graph Conversion Bot
- Client Organization: The Digital Forge, NYU School of Professional Studies
- Client Sponsoring Professional and Title: Dr. Andres Fortino, Professor of Management Systems & Director of The Digital Forge
- Project Goal: To design and develop an **automated RDBMS-to-Knowledge Graph conversion tool** that:
  - Automatically analyzes relational database schemas
  - Transforms them into graph data models (Neo4j)
  - Migrates data while preserving **referential integrity**
  - Enables **natural language querying (Text2Cypher)**

What did the project needed to achieve?

- Project Objective 1: Establish clear and approved project requirements
- Project Objective 2: Deliver a functional end-to-end prototype tool
- Project Objective 3: Validate correctness and reliability of the converted graph data
- Project Objective 4: Demonstrate usability through graph querying and documentation

What needed to be done?

Defined system requirements and project scope

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Designed system architecture and mapping logic

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Developed core system modules

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Performed data migration and validation

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Implemented query interaction layer

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Tested and evaluated system performance

Results

Schema Analyzer	Mapping Engine	Migration Engine	Validation	Query Interface
Key Capabilities				
• Connect to RDBMS • Extract schema metadata	• Rule-based mapping • Define node & relationship • **Human in-the-loop**	• Generates Cypher statements • Loads into Neo4j	• Completeness • Correctness • Detect duplicates	• Natural language input • Transfer to cypher
LLM integration				
	• Semantic interpretation			• NL → Cypher translation

- The data sets created for this project and the code for the tool may be found at: https://github.com/xjyjojo3845-dot/RDBMS-KnowledgeGraph-ConversionBot
- The demo of the tool: http://101.37.116.196:8501/

Quantitative Results

Dataset	Tables Processed	Node Types Generated	Relationship Types Generated	Nodes Created	Relationships Created	Unmatched Cases	Queries Tested	Queries Successfully Executed
Teaching Database	6	4	5	18	35	0	30	28
Chinook Database	11	9	9	4652	22289	0	30	26

- **100% data integrity preserved with zero unmatched cases across datasets.**
- **High successful rate for the NL question interpretations.**

Qualitative Results

1. Graph vs SQL: Key Insight

Sample Question: "Which playlists contain tracks from genres that Luís Gonçalves has purchased?"

- SQL:
  - Multiple JOINs
  - Complex and hard to read

```sql
SELECT DISTINCT p.Name
FROM Customer c
JOIN Invoice i ON c.customer_id = i.Customer_Id
JOIN invoice_line il ON i.Invoice_Id = il.Invoice_Id
JOIN Track purchased_track ON il.Track_Id = purchased_track.Track_Id
JOIN Genre g ON purchased_track.Genre_Id = g.Genre_Id
JOIN Track playlist_track ON g.Genre_Id = playlist_track.Genre_Id
JOIN playlist_track pt ON playlist_track.Track_Id = pt.Track_Id
JOIN Playlist p ON pt.Playlist_Id = p.Playlist_Id
WHERE c.First_Name = 'Luís' AND c.Last_Name = 'Gonçalves';
```

- Cypher: match the relationship logic

```cypher
MATCH (c:Customer {first_name: 'Luís', last_name: 'Gonçalves'})-[:PLACED]->(:Invoice)-[:CONTAINS]->(t:Track)-[:BELONGS_TO]->(g:Genre)
MATCH (p:Playlist)-[:CONTAINS]->(:Track)-[:BELONGS_TO]->(g)
RETURN DISTINCT p.name AS playlist_name
```

Graph queries are significantly more intuitive and expressive for relationship-heavy data.

2. LLM Insight

- Improves semantic interpretation
- Requires control and validation

Conclusions and Recommendations

Project success

Manual Migration → **Automated Pipeline**

Complex SQL → **Intuitive Graph Queries**

Limited Access → **Natural Language Interaction**

Next steps

- Improve **schema generalization** for different database structures
- Expand the **validation layer**
- Enhance **Text2Cypher accuracy and flexibility**

Thank you

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